

Libraries, Modules & Text as Data

Post-Assessment 5 · 11 questions · 39 points

Name: _____

Date: _____

Score: / 39

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Three ways to import

3 points

Write the line of code that uses each import to do the job.

Import	Job	Your line of code
<code>import string</code>	get all punctuation marks	
<code>from math import sqrt</code>	square root of 25	
<code>import random as r</code>	random number 1 to 6	

2. kidtools.py

3 points

FILE: KIDTOOLS.PY

```
def double(x):
    return x * 2
```

FILE: MAIN.PY

```
import kidtools as kt
from kidtools import double
print(kt.double(21))
print(double("ab"))
```

(a) What does `main.py` print?

(2 pts)

(b) What must the module file be named exactly for `import kidtools` to work?

(1 pt)

3. Library words

4 points

Meanings:

1. A single `.py` file of functions and classes.
2. A folder of modules that belong together.
3. An online collection of free Python libraries anyone can install.
4. The tool that downloads and installs third-party libraries.
5. A big collection of code for one purpose, like `matplotlib`.

(a) Write the number of the matching meaning next to each word.

(3 pts)

Word	Meaning number
library	
module	
pip	

(b) Circle every library below that you must install with pip before you can import it: `math`

(1 pt)

`matplotlib` `random` `rich`

4. Read the docs

3 points

FROM THE PYTHON DOCS

```
str.center(width, fillchar=' ')\n    Return the string centered in a string of length width.\n    Padding is done using fillchar (default is a space).
```

(a) What is the smallest number of arguments you must give?

(1 pt)

(b) What is `fillchar` if you leave it out?

(1 pt)

(c) Write a call that centers `"Hi"` in a string of length 10, padded with `*`.

(1 pt)

5. Turtle Art Studio

4 points

```
import turtle
pen = turtle.Turtle()
for i in range(6):
    pen.forward(50)
    pen.left(60)
```

(a) What shape does this draw?

(1 pt)

(b) Which two numbers would you change, and to what, so it draws a pentagon (5 sides)? (2 pts)

(c) `forward` is a _____ of the `Turtle` object.

(1 pt)

6. Class Survey Dashboard

4 points

Fill in the four blanks to draw a labeled bar chart.

```
import matplotlib.pyplot as plt
sports = ["soccer", "swim", "chess"]
counts = [15, 6, 8]
plt.bar(_____, _____)
plt.title("Favorite Sports")
plt._____("Sport")
plt.ylabel("Students")
plt._____()
```

7. Loading & Cleaning Text

4 points

```
import string
text = "Go, dog. Go! Big dog, go."
for p in string.punctuation:
    text = text.replace(p, "")
words = text.lower().split()
print(words)
print(len(words), len(set(words)))
```

(a) Write what the first print shows.

(2 pts)

(b) Write what the second print shows.

(1 pt)

(c) Lexical diversity = unique words ÷ total words. What is it for this text? (A fraction is fine.)

(1 pt)

8. Top Words

4 points

```
from collections import Counter
words = ["a", "ship", "and", "a", "storm", "ship",
        "a", "sea", "ship", "a", "storm"]
stop_words = {"the", "and", "a"}
counts = Counter(words)
print(counts["ship"])
print(counts.most_common(1))
kept = [w for w in words if w not in stop_words]
print(Counter(kept).most_common(2))
```

(a) Write exactly what the program prints.

(3 pts)

(b) Why should you use `.lower()` on the text before counting words?

(1 pt)

9. Author Fingerprint

3 points

```
poem1 = {"moon", "star", "sun", "cloud"}
poem2 = {"sun", "rain", "cloud", "wind", "snow"}
```

Write the words in each result (any order), or the number.

Expression	Result
poem1 & poem2	
poem2 - poem1	
len(poem1 poem2)	

10. Bug Safari

2 points

```
print("Stop, thief! I'm fast.".split())
```

(a) What does this print?

(1 pt)

(b) Why is this a problem if we want to count words?

(1 pt)

11. Book Detective

5 points

Write a function `lexical_diversity(filename)` that:

- opens the text file; if it is missing, prints `File not found` and returns 0
- makes the text lowercase and removes punctuation (use `string.punctuation`)
- splits it into words and **returns** unique words ÷ total words

EXAMPLE: FOR A FILE CONTAINING "GO, DOG. GO! BIG DOG, GO." IT RETURNS

0.5

END OF ASSESSMENT